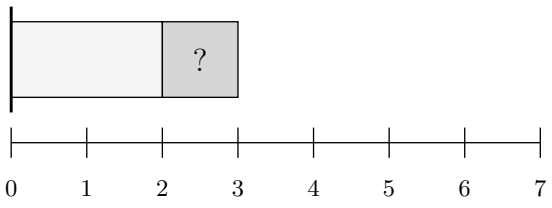
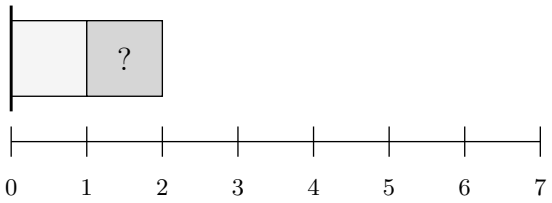
A hidden end can be anywhere in its gray band, including the edges. Each trial may use new lengths. Join strips end to end from 0, with no gap or overlap.

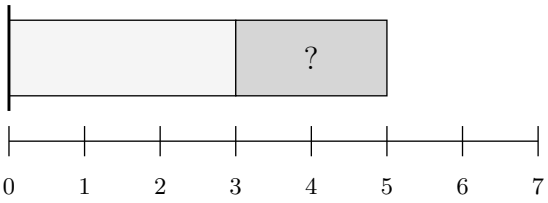
A: 2 to 3 units

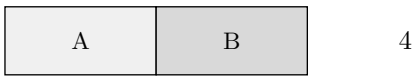


B: 1 to 2 units



A + B: 3 to 5 units

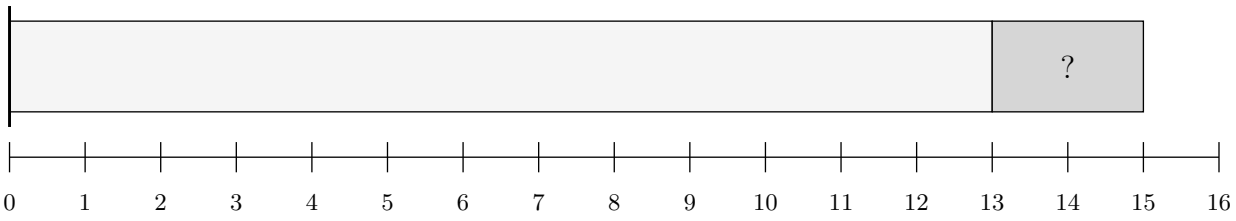




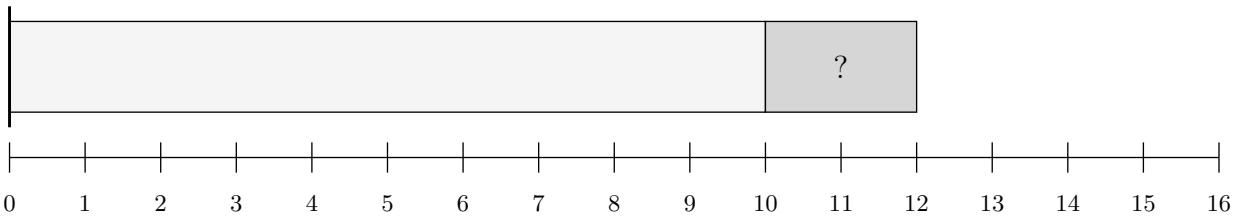
4

Problem 5: Align both starts at 0. Find the tightest guaranteed range for the gap between the ends. Build lengths that show neither bound can be improved.

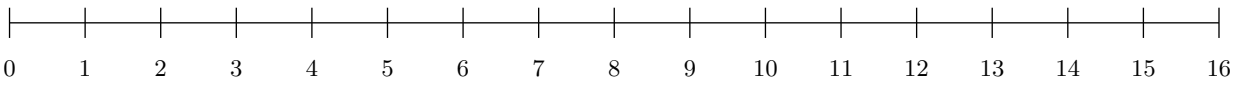
long strip: 13 to 15 units



short strip: 10 to 12 units

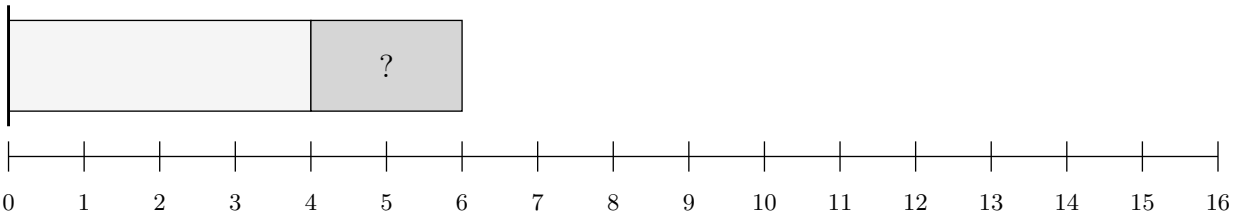


Problem 6: Someone subtracts the two middle values and announces one exact gap. Can you choose allowed strips that show why this is unsafe?

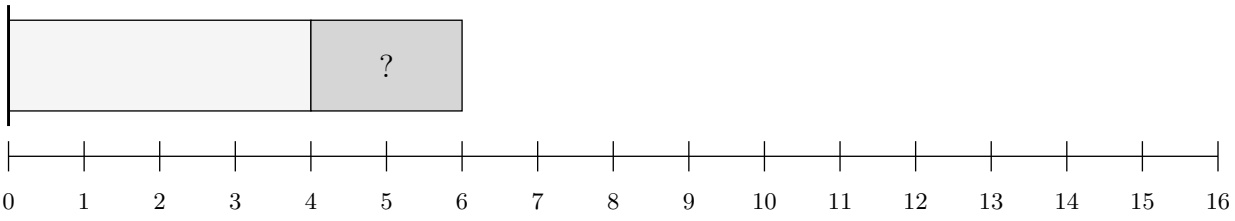


Problem 7: Compare the gaps between the right ends of these joins, starting at 0. Repeated A means the same unchanged length; B may vary separately. Find the tightest range for each gap.

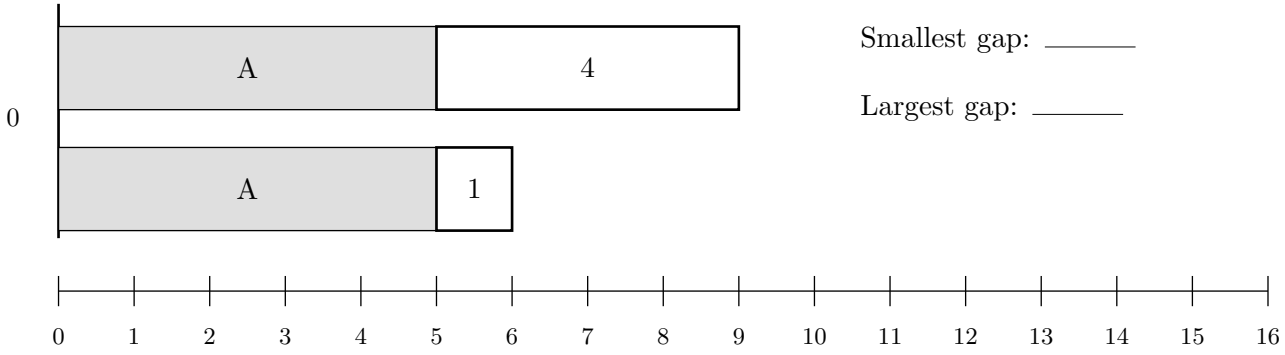
A: 4 to 6 units



B: 4 to 6 units

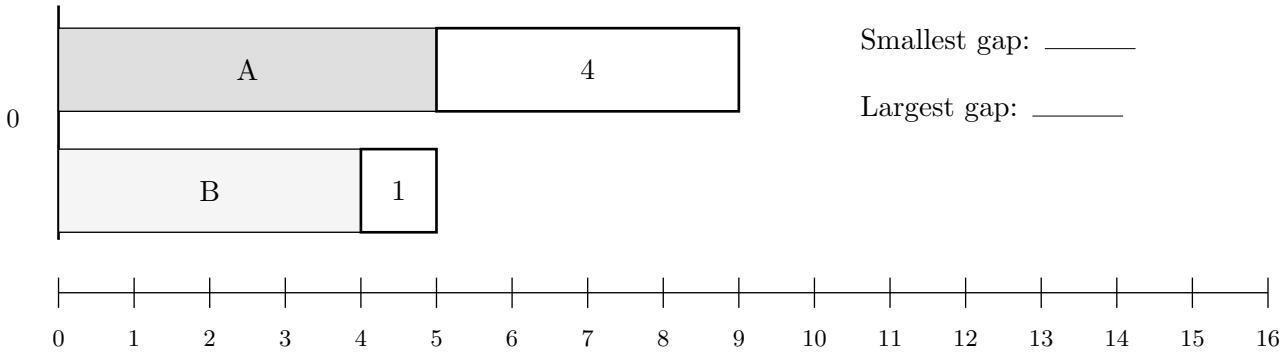


The white pieces are exactly 4 and 1 units long.



Smallest gap: _____

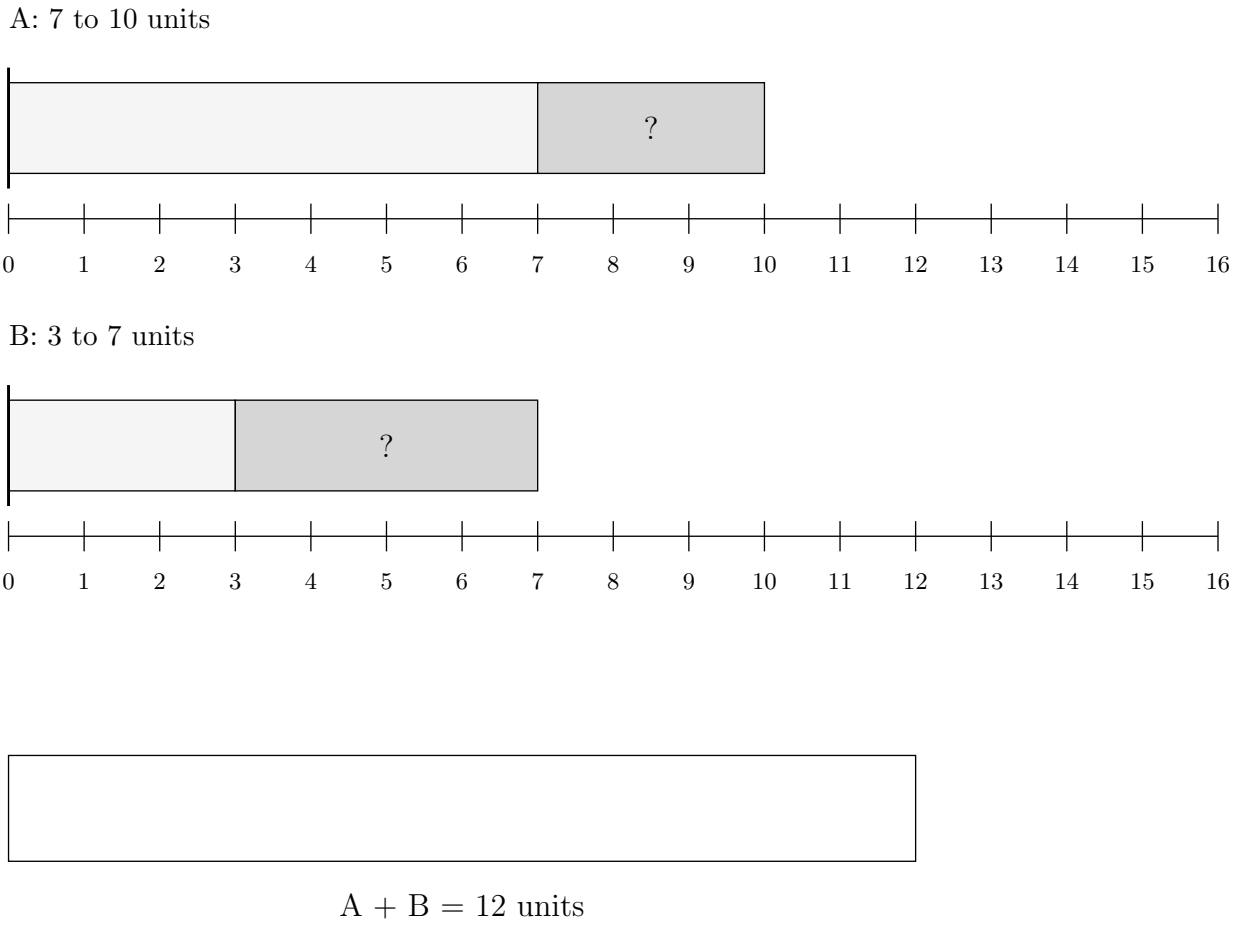
Largest gap: _____



Smallest gap: _____

Largest gap: _____

Problem 8: A is 7 to 10 units and B is 3 to 7 units. You learn that their join is exactly 12 units. Align their starts at 0. Find the smallest and largest possible gap between their ends.



Problem 9: Join A (4 to 6 units) to B (3 to 4 units), then remove that very same A. Someone uses the join's range and A's range separately to claim the remainder could be 1 to 6 units. Is that the tightest range? Explain with the pieces.